

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Pharm. Examination

University Theory Examination November-December 2015

PH205 Pharmaceutical Engineering-III

Date: 01/12/2015, Tuesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following.

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|---|---|----|
| A | 1) Enumerate the continuous process crystallizers. Explain in detail swenson-walker crystallizer. | 05 |
| | 2) Describe the basic functions and elements of automatic process control system. | 05 |
| B | 1) Enumerate the methods of size separation and explain cyclone separator. | 05 |
| | 2) Describe the process of nucleation. What are the factors that affect nucleation? | 05 |
| C | 1) Write note on size separation by elutriation. | 05 |
| | 2) Write note on methods to achieve supersaturation. | 05 |

Q 2 Attempt any TWO of the following.

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|---|---|----|
| A | 1) Write note on feed-forward Automated Process Control System. | 05 |
| | 2) Write note on standards for sieve. | 05 |
| B | 1) Write note on application of size separation in pharmacy. | 05 |
| | 2) Define Crystallization. Explain objectives of crystallization. | 05 |
| C | 1) Write note on application of crystallization in pharmacy. | 05 |
| | 2) Write a note on standards for powder. | 05 |

SECTION- II

Q 3 Attempt any TWO of the following.

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|---|---|----|
| A | Explain factor affecting size reduction and give the brief note on modes of motion in size reduction. | 10 |
| B | Classify size reduction equipments and explain Tumbling mill in detail. | 10 |
| C | Explain Theory of Mixing. Enumerate types of mixtures and explain planetary mixture in detail. | 10 |

Q 4 Attempt any TWO of the following.

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|---|--|----|
| A | Explain in detail the methods for waste water treatment in industries. | 10 |
| B | Explain chemical hazards and give the detail on fire hazards with its prevention. | 10 |
| C | Give the Principle, construction, working, uses, advantage, disadvantage and variants of micronizers with neat and clean diagram | 10 |